



## Contact

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Hyderabad, Telangana, India



## Education

**M.Tech in Electronics & Communication Systems,**

AITS(JNTUH) Hyderabad|

2011-2013

**Marks 71%**

## EXPERIENCE

**MAURYA KAUTILYA Harmoniques Pvt Ltd/Board of Director Hyderabad | 5th march 2025 to till date**

1. Developed a series of for high power amplifiers for a Space. Resulting in a reduction of production cost.
2. Conduct the extensive simulations & analysis to ensure that the RF design met the product requirements.
3. Utilized advanced Techniques in RF design to reduce the size & cost of the system.

**Astra Microwave Products Ltd | Team Lead Hyderabad | July 2022 – 5th march 2025**

1. Designed, developed, and executed RF test setups for validation and qualification of L, S, and C-band payload subsystems for ISRO satellite applications.
2. Performed S-parameter measurements, power testing, spurious and IMD characterization, and phase noise analysis using Vector Network Analyzer (VNA) and Spectrum Analyzer.
3. Performed RF tuning and optimization for modulator output power, phase balance, and spectral purity.
4. Conducted end-to-end RF link validation between S-Band uplink and C-Band downlink subsystems.
5. Conducted Noise Figure, Gain, and Linearity measurements using Noise Figure Analyzer and Power Meter. Verified BOMs, schematics, and test documentation for MMIC-based RF modules, ensuring design compliance and manufacturability.



## Professional Summary

A brief overview that highlights your most relevant qualifications, leadership experience, and achievements to show you are a strong candidate for a director role..



## Professional Experience

**TKR College of Engineering & Technology | Assistant Professor** Hyderabad | 2012 – 2022

- 1.Delivered undergraduate courses in Fundamentals of Electrical Engineering, Digital Logic Design, C Programming, FPGA Programming, and Verilog HDL.
- 2.Designed and implemented practical lab experiments to reinforce theoretical learning in digital and embedded systems.
- 3.Guided students in mini and major projects involving digital design using Verilog and FPGA-based applications.
- 4.Conducted hands-on lab sessions using tools such as Xilinx ISE / Vivado, ModelSim, and Keil Vision. Mentored students on logic synthesis, timing analysis, and RTL coding for FPGA-based systems. Developed course materials, assignments, and assessment modules following JNTU Hyderabad curriculum standards.
- 5.Collaborated with faculty teams to develop academic projects integrating RF and digital subsystems for real-world applications.
- 6.Evaluated student



## Projects / Open-Source

### 1. Direct Digital Modulator (DDM) IRNSS Payload(SAC)

Contributed to the development and qualification of L1, L5, and S-Band DDM units for IRNSS navigation satellites. Performed RF tuning and optimization for modulator output power, phase balance, and spectral purity. Developed FPGA based control modifications and integrated with RF hardware for real-time data modulation. Conducted environmental testing (vibration, thermovac) and supported qualification & light model validation. Utilized VNA, Spectrum Analyzer, and DSO for performance verification and compliance testing.

### 2. Multi-Object Tracking Radar (MOTR)(SHAR-SDSC)

Supported testing of mmWave radar subsystems for multi-target detection and tracking. Performed TRM (Transmit/Receive Module) and TRU controller card validation and calibration. Conducted functional and RF performance testing using VNA and spectrum analysis tools. Debugged and optimized RF signal chains for improved tracking accuracy and sensitivity. Prepared detailed test reports and validation data for customer review and documentation.

## Design Skills

- **Programming Languages:** C Programming, Verilog, Embedded C, FPGA Programming
- **Tools & Technical skills:** Xilinx ISE, Vivado, ModelSim, Keil Vision, OrCAD, Multisim, LabView, Vector Network Analyzer, Spectrum Analyzer, RF Tuning, Trouble shooting, Signal Processing.
- **Leaderships & Soft Skills:** Emphasize leadership, communication, problem-solving, analytical thinking, and project management. Focus on director-level responsibilities like team leadership, project management (budget, timelines), and strategic direction, supported by a strong technical foundation in RF design, testing, and troubleshooting.

### 3. Mobile Satellite System (MSS) Terminal(SAC)

Supported design, development, and testing of S-Band mobile satellite terminals for unmanned railway level-crossing warning systems. Conducted end-to-end RF link validation between S-Band uplink and C-Band downlink subsystems. Executed RF alignment, tuning, and power calibration for MSS terminal transmit/receive modules. Assisted in hardware verification using OR-CAD schematics and system-level test documentation.

### 4. Wideband Tuner with LO (PLL)(BEL)

Contributed to development of a 1.22.4 GHz PLL-based local oscillator for wideband receiver/transceiver applications. Validated VCO phase noise performance and PLL lock stability across temperature ranges. Executed frequency synthesizer tuning, output leveling, and noise pure optimization. Collaborated on hardware debugging of PLL and VCO boards with STM-32 control interface. Documented test procedures, failure analysis, and re-test philosophy for qualification units.

### 5. C-Band 15 W SSPA with EPC(SAC)

Performed testing, tuning, and characterization of 15 W C-Band SSPA units with integrated EPC (Electronic Power Conditioner). Validated gain, efficiency, linearity, and output power of SSPA across operating frequency range. Conducted load-pull and thermal tests to ensure space-grade performance compliance. Supported production and qualification testing for GEOSAT series payloads. Collaborated with design and production teams for indigenous SSPA technology transfer and process optimization.

### 6. Ocean-SAT 3A(URSC)

A CDMA receiver demodulates a signal by multiplying the received signal with a unique spreading code that matches the one used by the transmitter. This process, known as despreading, extracts the original data from the combined signal, which also contains signals from other users, by correlating the desired signal with its specific code. The receiver uses a synchronization loop to align the local code replica with the incoming signal, and a RAKE receiver may be used to combat multipath fading.

### Honors & Awards

GATE (Graduate Aptitude Test in Engineering) Qualified in 2011 (ECE Discipline) RAC / DRDO External Expert Examination Qualified in 2018 (Government of India Defence Undertaking)

### Declaration

I hereby declare that the information provided above is true to the best of my knowledge.

**Date:**

**Place:** Hyderabad

**Rushi N**

